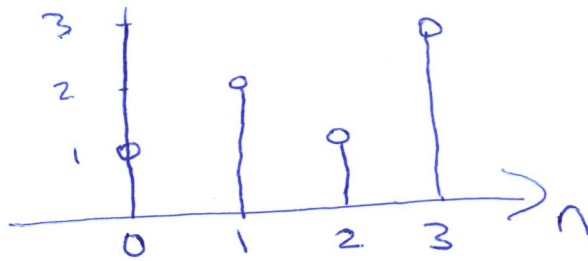


1) a)



$$b) \quad y[n] = \frac{1}{2} (x[n] + x[n-1])$$

$$y[0] = \frac{1}{2} (x[0] + x[-1]) = \frac{1}{2} (1 + 0) = \frac{1}{2}$$

$$y[1] = \frac{1}{2} (x[1] + x[0]) = \frac{1}{2} (2 + 1) = 1\frac{1}{2}$$

$$y[2] = \frac{1}{2} (x[2] + x[1]) = \frac{1}{2} (1 + 2) = 1\frac{1}{2}$$

$$y[3] = \frac{1}{2} (x[3] + x[2]) = \frac{1}{2} (3 + 1) = 2$$

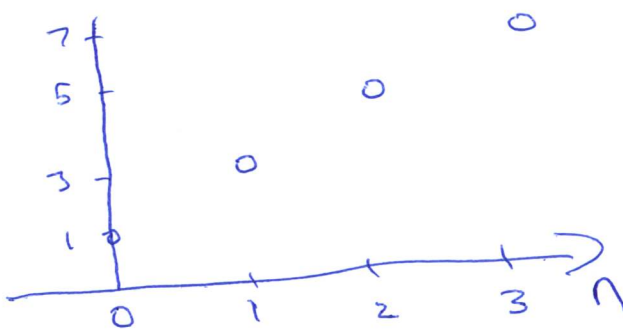
$$y[4] = \frac{1}{2} (x[4] + x[3]) = \frac{1}{2} (0 + 3) = 1\frac{1}{2}$$

c) 1 sample

d) Variance (σ^2) reduction by $\frac{1}{M}$ So standard deviation (σ) = $\frac{1}{\sqrt{M}}$ M = 2 so σ reduction by $\frac{1}{\sqrt{2}}$

e) 1 sample

2) a)



2b)

$$y[n] = x[n] - x[n-1]$$

$$y[0] = x[0] - x[-1] = 1 - 0 = 1$$

$$y[1] = x[1] - x[0] = 3 - 1 = 2$$

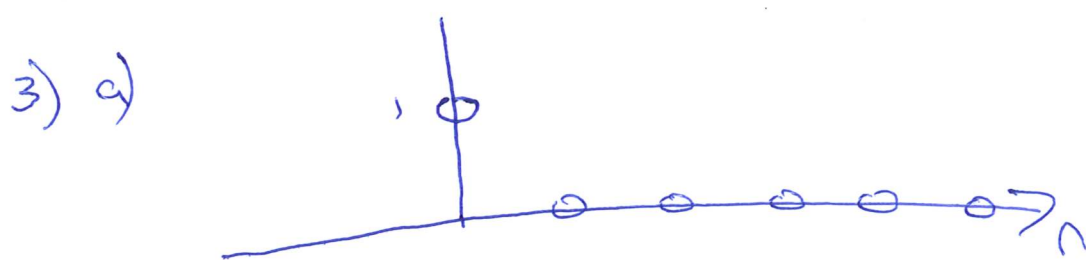
$$y[2] = x[2] - x[1] = 5 - 3 = 2$$

$$y[3] = x[3] - x[2] = 7 - 5 = 2$$

$$y[4] = x[4] - x[3] = 0 - 7 = -7$$

c) 1 sample

d) Differentiator



b)

$$y[n] = 0.5y[n-1] + 0.5x[n]$$

$$y[0] = 0.5y[-1] + 0.5x[0] = 0 + 0.5 = 0.5$$

$$y[1] = 0.5y[0] + 0.5x[1] = 0.5 \times 0.5 + 0 = 0.25$$

$$y[2] = 0.5y[1] + 0.5x[2] = 0.5 \times 0.25 + 0 = 0.125$$

$$y[3] = 0.5y[2] + 0.5x[3] = 0.5 \times 0.125 + 0 = 0.0625$$

$$\vdots$$

c) It is the response to a discrete-time impulse function.

4) a) Let $w[n] = ax[n] + by[n]$

$$\begin{aligned} W(z) &= \sum_{n=-\infty}^{\infty} w[n] z^{-n} \\ &= \sum_{n=-\infty}^{\infty} (ax[n] + by[n]) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} ax[n] z^{-n} + \sum_{n=-\infty}^{\infty} by[n] z^{-n} \\ &= a \sum_{n=-\infty}^{\infty} x[n] z^{-n} + b \sum_{n=-\infty}^{\infty} y[n] z^{-n} \end{aligned}$$

$$W(z) = aX(z) + bY(z)$$

b) Let $w[n] = x[n-1]$

$$\begin{aligned} W(z) &= \sum_{n=-\infty}^{\infty} w[n] z^{-n} \\ &= \sum_{n=-\infty}^{\infty} x[n-1] z^{-n} \end{aligned}$$

Let $m = n-1$, so $n = m+1$

$$\begin{aligned} W(z) &= \sum_{m=-\infty}^{\infty} x[m] z^{-(m+1)} \\ &= z^{-1} \sum_{m=-\infty}^{\infty} x[m] z^{-m} \end{aligned}$$

$$= z^{-1} X(z)$$

if $x[-1] = 0$
i.e. causal.

c) $y[n] = 0.7x[n] + 0.3x[n-1]$
 $Y(z) = 0.7X(z) + 0.3X(z)z^{-1} = (0.7 + 0.3z^{-1})X(z)$

d) $y[n] = 0.7y[n-1] + 0.3x[n]$
 $Y(z) = 0.7Y(z)z^{-1} + 0.3X(z)$
 $Y(z)(1 - 0.7z^{-1}) = 0.3X(z)$
 $Y(z) = \frac{0.3}{1 - 0.7z^{-1}} X(z)$